

Arrakis

A 3D Scene of Spice Harvester in Arrakis Dune

by

Asique Ehetasamul Haque (2107096)

Description

The project is an interactive 3D scene that will be developed in C++ and OpenGL, featuring a massive Spice Harvester navigating the deep desert. This is an imaginary scenery from Frank Herbert's novel series 'Dune'. The scene includes a detailed harvester with rotating tank treads, an Ornithopter observing the dunes and the harvester, a rhythmic thumper device actively pounding the sand, a procedurally generated terrain of rolling sand dunes, drifting clouds of orange spice dust, and a harsh, sun-baked lighting environment. An on-screen dashboard displays live machinery status, thumper frequency, and spice collection metrics.

Objects in the Scene

- A massive Spice Harvester built with a detailed chassis, a dark-glass cockpit, and rolling tank treads.
- A massive Ornithopter observing the Spice Harvester from atop the desert dunes.
- A Thumper device anchored in the foreground, featuring a stationary metallic base and a continuously oscillating central piston.
- A procedural ground plane modeled as a high-density grid mesh, manipulated by mathematical functions to create rolling sand dunes and textured with a gritty surface.
- Drifting spice dust (alpha-blended flattened planes or particles) hovering around the harvester's tracks to simulate disturbed sand.
- A black, smokey screen of clouds moving upwards, depicting the active operational mode of the Spice Harvester.
- A dashboard UI will display live machinery status, thumper frequency, and spice collection metrics.

Features

Transformations: Translation (harvester moving linearly across the desert, dust particles drifting horizontally, thumper piston translating vertically); Rotation (tank treads continuously spinning); Scaling (dust particles shrinking/growing as they dissipate); Hierarchical (harvester chassis → treads; thumper base → oscillating piston).

Animation: Continuous motion across all elements above: the slow crawl of the harvester, the sharp mechanical pounding of the thumper, and the smooth drift of the spice dust.

Coordinate System and Projection: Y-axis = altitude (dune height), X-axis = horizontal desert expanse, Z-axis = depth; Perspective projection for the immersive 3D scene, Orthographic projection for the dashboard HUD.

Camera: Cinematic low-angle Overview, Follow (locked to the harvester), Free-fly, and Cockpit POV modes; arrow keys move/orbit the free camera.

Lighting: Ambient + Diffuse + Specular (base Phong model); Directional (the distant Arrakis sun casting parallel, warm light); Spotlight (a vibrant orange under-glow mounted beneath the harvester, pointing directly down at the sand).

Shadow: Dynamic, sharply defined shadows cast by the harvester and the thumper onto the irregular geometry of the sand dunes, calculated based on the angle of the directional sun.